Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 5 times with different random seeds. The learning rate was set to 0.0003 with the Adam optimizer.

2 Results

Figure 1 shows the test accuracy of CNN by dataset.

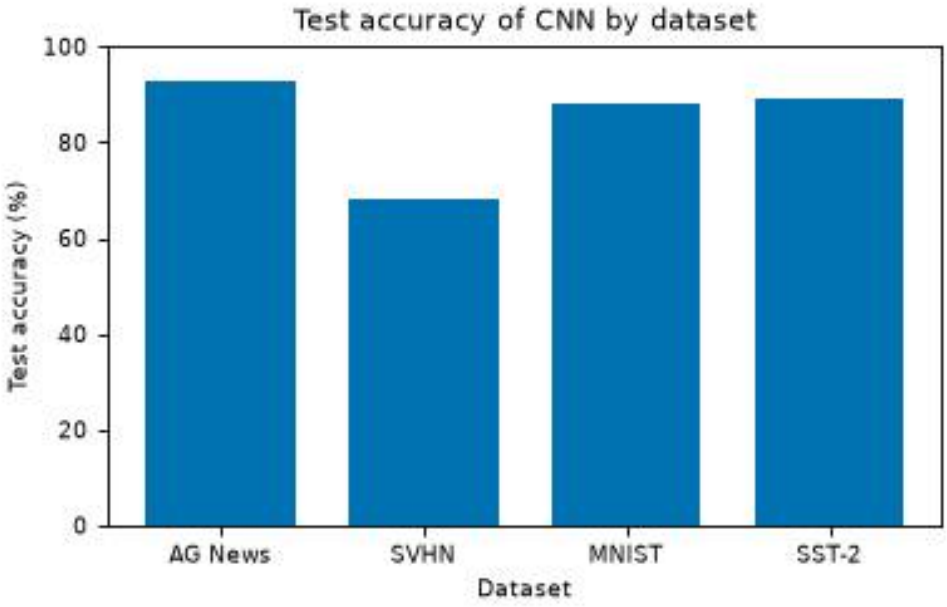


Figure 1: Test accuracy of CNN by dataset for the models in this experiment.

On SST-2, CNN obtains a test accuracy of 88.9%. As shown in Figure 1, CNN achieves 88.9% test accuracy on SST-2. CNN scores 54.1 points higher on AG News than on SST-2.

Figure 2 shows the validation accuracy during training.

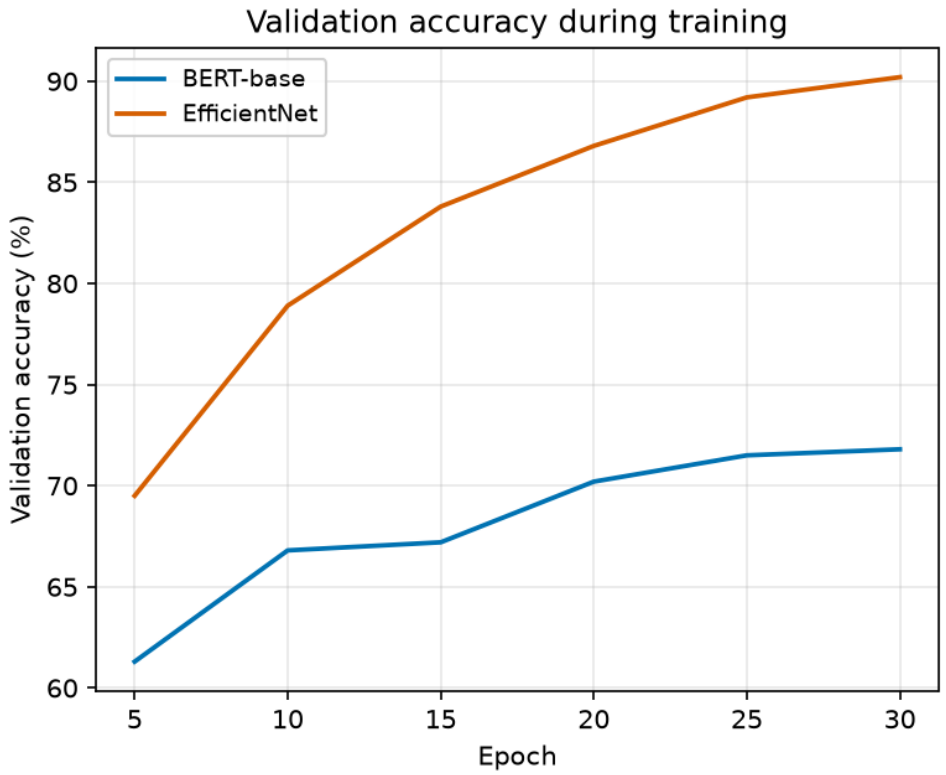


Figure 2: Validation accuracy during training for the models in this experiment.

BERT-base reaches 90.2% validation accuracy at epoch 30. As shown in Figure 2, at epoch 10, EfficientNet is 12.7 points better than BERT-base.

Figure 3 shows the test accuracy by dataset.

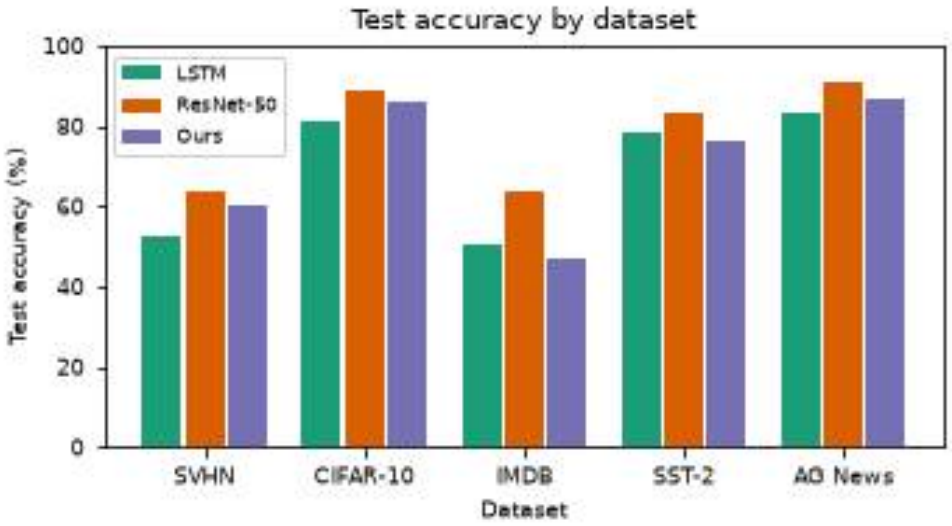


Figure 3: Test accuracy by dataset for the models in this experiment.

On AG News, ResNet-50 is 8.0 points better than LSTM. As shown in Figure 3, ResNet-50 reaches 78.3% test accuracy on SST-2. On CIFAR-10, ResNet-50 obtains a test accuracy of 69.1% (Figure 3). As shown in Figure 3, LSTM achieves 63.6% test accuracy on SVHN.

3 Discussion

Naive Bayes reaches 90.2% accuracy on IMDB. We note that the training was stable for all models.